Session 4: Technical Drawings, Exploded View and Preliminary BOM

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This report details the technical architecture, component specifications, and manufacturing requirements for the UnBROKEN Skate Wrap, a dual-function tool organizer and hands-free skateboard transport system.

3.1 Product Architecture
The architectural framework of the UNBROKEN Skate Wrap relies on a specialized, adaptive folding geometry. Rather than following the rigid spatial layout of a traditional backpack, this system is engineered around a dynamic "wrap" mechanism. When unbuckled and fully unfolded, the architecture behaves as a completely flat, multi-compartment internal tool tray designed to facilitate immediate, on-street skate maintenance. When closed, it physically wraps around and encloses a standard skateboard deck for secure transit.

To optimize manufacturing and assembly workflows, the product layout is divided into three distinct functional layers:

1. Protective Outer Layer (Sacrificial Surface): Positioned on the exterior envelope, this architectural zone is designed as a rugged shield to directly bear the intense, destructive friction caused by the 80-grit silicon carbide abrasive griptape of the skateboard.

2. Internal Modular Organizer Tray: The primary user-facing touchpoint when the system is flat. This zone features texturized, flexible fabric tailored with dedicated, elastic-retention slots and airtight cells to organize heavy tools, bearings, wheels, and volatile chemical compounds like skate wax.

3. Ergonomic Suspension Zone: A heavy-duty, load-bearing strap interface. This structure is strategically engineered to suspend, distribute, and stabilize the cumulative weight of both the board and the tools comfortably across the user's upper torso during multi-modal urban commutes.

3.2 Parts Breakdown

Manufactured Parts

These components require custom tooling, specific fabric patterns, and dedicated processing operations by an OEM or supplier:

- Main Body / Outer Shell Face: The structural footprint cut to accommodate an 85 cm deck.
- Internal Organizer Canvas Tray: Multi-paneled soft-goods assembly with tool paths.
- Airtight Tool Pockets: Specially laminated polymer cells for leak-proof wax containment.
- Back-Panel Cushioning: Ergonomic foam padding protecting the lower back.
- Cultural Identity Accents

Commercial Parts (Off-The-Shelf / OTS)

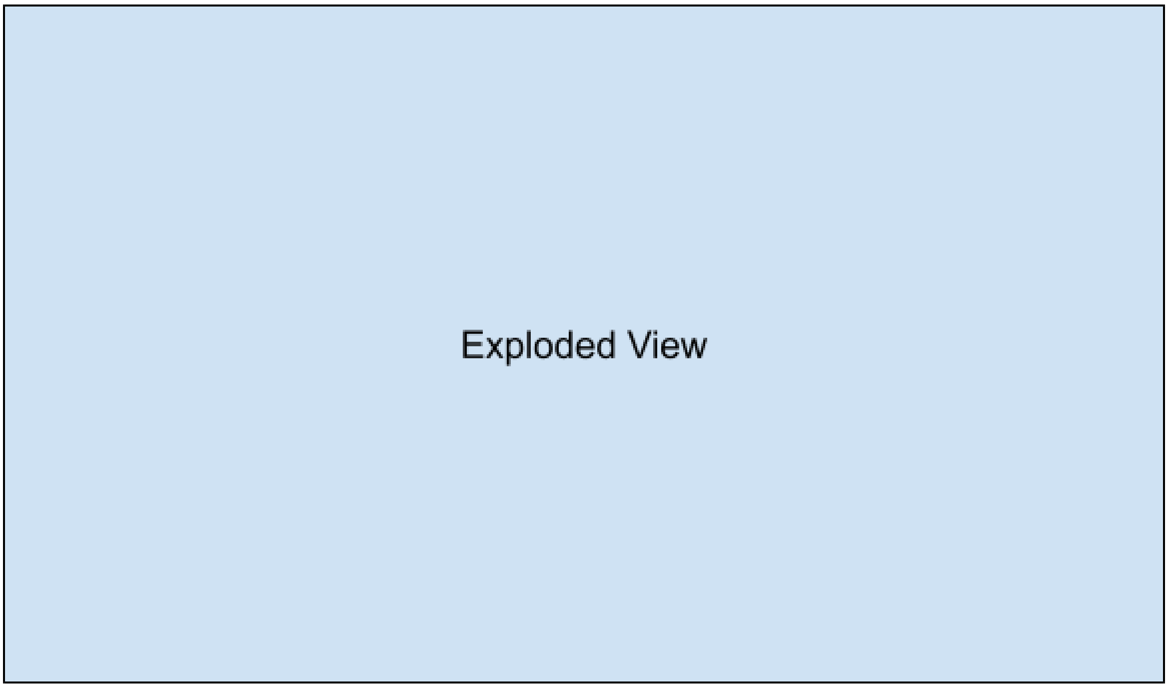
Standard, highly reliable industrial hardware sourced from specialized distributors to reduce upfront tooling costs:

- Reinforced Nylon Webbing: High-tensile, industrial-grade flat webbing.
- Side-Release Buckles: High-impact Polyoxymethylene (POM / Acetal) female/male quick-clip couplers.
- Ladderlock Tension Sliders: POM dual-slot adjusters utilized for precision strap cinching.

3.3 Exploded View Description

The engineering exploded view visualizes the multi-layered assembly of the soft-goods product. At the base layer sits the Main Outer Shell (woven from high-modulus Aramid or UHMWPE fibers), which acts as the core chassis. Suspended directly above it is the Back-Panel Padding Core, composed of high-density closed-cell EVA foam. This core is structurally sandwiched beneath the Internal Organizer Tray (made of durable stretch denim or technical texturized nylon). This stacking ensures the foam acts as a physical buffer absorbing structural vibration from the board’s metal trucks and hard wooden deck edges.

At the distal boundaries, the hardware anchors are exposed. The two adjustable straps calculated mathematically at an ergonomic length of 64 cm are displayed exploding outward from high-stress anchor coordinates on the outer shell. The POM side-release and ladderlock buckles are mapped inline with the nylon webbing terminal ends to show the clear fastening sequence required to secure the overall wrap configuration around a skateboard.



3.4 Drawing Checklist

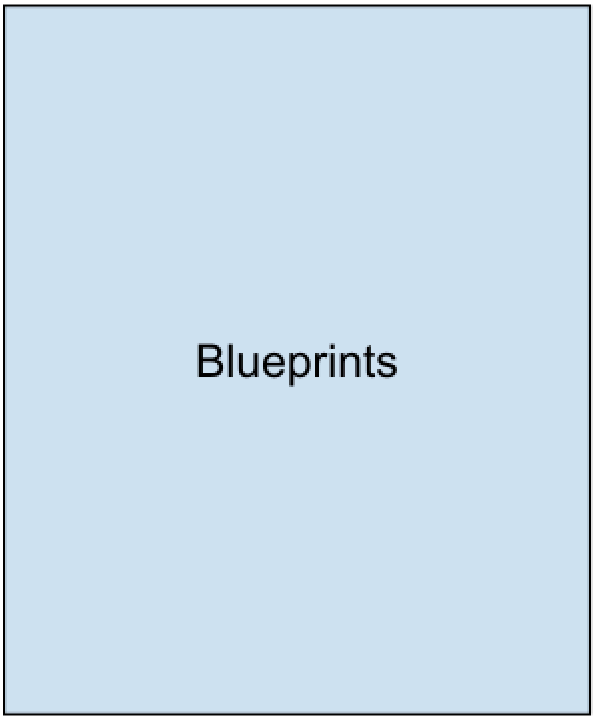
To transition the Skate Wrap from a conceptual stage to industrial factory tooling, a formalized engineering drafting package must be completed, verifying compliance with mechanical safety benchmarks (specifically EN388 and EN420 standards):

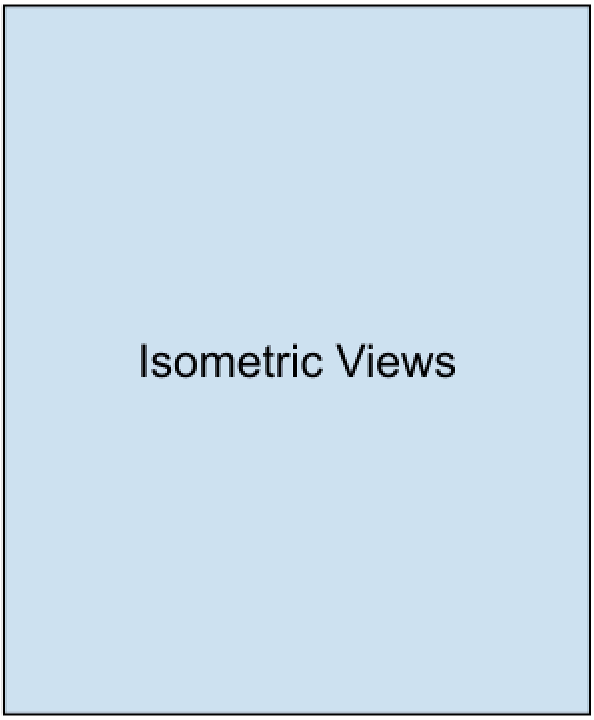
- Isometric Views: Comprehensive, dimensioned isometric drawings visualizing the exterior volume in both its unrolled/flat "tray" mode and its active, tightly bound "wrap" transport mode.

- Exploded Assembly Diagram: A complete multi-axis assembly blueprint explicitly mapping every independent layer, stitch zone boundary, and hardware fastening point.

- Technical Cross-Sections: Precise cross-sectional cuts mapping the thickness of the internal closed-cell EVA foam core and verifying the continuous boundary edges of the PU/TPU leak-proof fluid barriers.

- Stitching & Bar-Tack Blueprint Detail: Tooling callouts illustrating the precise spacing, pattern directions, and densities for both the manual lockstitches and the heavy machine bar-tack anchors.





3.5 Preliminary Bill of Materials (BOM)

Part / Component	Quantity	Type	Material	Process or Supplier	Estimated Cost	Function	Status
Main Body / Outer Shell	1	Manufactured	Aramid (Kevlar) or UHMWPE reinforced Nylon	Machine Sewing / UV Grafting	To be defined	Griptape/ Abrasion protection	Needs testing
Internal Organizer Tray	1	Manufactured	Stretch Denim / Nylon	Computerized Lockstitch	To be defined	Tool organization / Branding	Ready for prototype
Transport Straps	2	Commercial	Reinforced Nylon Webbing	Cutting + Sewing	To be defined	Support board weight	Ready for prototype
Connectors (Buckles)	Needs measurement	Commercial	High-impact POM / Acetal	Injection / CNC	To be defined	Quick-strap access	Check tolerance
Back Panel Padding	1	Manufactured	High-density EVA or PU Foam	Foam/Textile Assembly	To be defined	Ergonomic buffer	Check cleaning
Airtight Pockets	Needs measurement	Manufactured	PU or TPU coated fabric	Hot-Melt Lamination	To be defined	Prevent wax/oil leaks	Needs evidence
Identity Accents	Needs measurement	Manufactured	Rattan fitrit / Batik / Tenun	Artisanal weaving	To be defined	Subcultural identity	Ready for prototype

3.6 Assembly Sequence

1. Polymer Membrane Lamination: Apply a liquid PU or TPU matrix directly onto the interior faces of the Nylon or Denim fabric utilizing a hot-melt lamination or knife-over-roll coating process to solidify the leak-proof pocket partitions.
2. Internal Tool Slot Assembly: Stitch down the elastic retention bands and layout folds onto the internal modular tray using industrial lockstitch machines.
3. Cushioning Integration: Position the high-density EVA foam core slice onto the back panel layout, sandwiching it between the internal canvas tray and the exterior backing structure.
4. Structural Framing Stitch: Join the high-performance outer aramid shell with the internal tray assembly using automated sewing setups running heavy-weight bonded nylon or aramid threads.

5. Load-Bearing Bar-Tack Reinforcement: Route the nylon webbing strap lines through the POM ladderlock sliders and permanently anchor them to the structural shell using a high-density bar-tack stitch matrix at maximum stress coordinates.
6. Artisanal Rattan Border Finish: Manually integrate the raw rattan fitrit structural strips along the terminal layout seams of the casing to serve as a physical frame and aesthetic accent.

3.7 Technical Risks

- Frictional/Abrasive Fatigue: Silicon carbide grains from the 80-grit griptape act as a constant grinding surface. Under regular cyclic vibration, the technical fibers will inevitably undergo micro-fraying, which limits the total life cycle of the outer shell textile face.

- BOM Cost Escalation vs. Subcultural Pricing Limits: Sourcing ultra-performance protective textiles like Kevlar® or UHMWPE heavily escalates manufacturing costs. If the final retail MSRP slides into a luxury tier, the core DIY-centric skate demographic will reject the product as inauthentic and revert to zero-cost, improvised alternatives.

- Chemical Degradation & Migration: Sourcing lower-cost PVC coatings carries the risk of plasticizer migration (such as DOP) when in direct contact with hydrocarbon skate waxes and petroleum lubricants, potentially melting or degrading the internal pockets over time.

3.8 Missing Information and Next Steps

Missing Information

1. Quantitative Mass Data: The exact weight calculation of the technical textile assembly has yet to be finalized to guarantee it does not add significant weight to the board.
2. Financial Target Matrix: Sourcing agreements and final factory production quotes are required to determine the retail target MSRP.
3. Polymer Compatibility Verification: Radio Frequency (RF) or thermal welding compatibility criteria between alternative TPU/EVA barriers need empirical verification.

Next Steps

1. Fabricate Final Prototyping Run: Construct a final, high-fidelity physical prototype utilizing the verified materials (Aramid/TPU/Denim) to run true structural and ergonomic usability tests.
2. Accelerated Environmental Lab Testing: Subject the synthetics to specialized lab ovens to observe thermal oxidative degradation, simulating long-term outdoor exposure to urban environments.
3. Interface Testing: Launch beta-testing protocols for the digital online configurator layout using a closed testing group composed of native skaters.
4. Legalize Subcultural Partnerships: Formalize commercial contracts and mutual alignment agreements with local pro skateboarders like Max Barrera and Itzel Granados ahead of the commercial release timeline.